



Explainable Multi-source AI Framework for Real-Time Stock Price Monitoring and Prediction

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Abstract

India's real-time forecasting of stock prices is hindered by heterogeneous data sources, volatility in markets, and regulatory demands for transparency in algorithmic trading. This research proposes a systematic, interpretable multi-source artificial intelligence (AI) framework for tracking and predicting stock prices on the NSE and BSE. The model brings together market technical indicators, macroeconomic variables, and sentiment factors into one prediction pipeline through a hybrid ensemble of temporal fusion transformers (TFTs), long short-term memory (LSTM) networks, random forests (RFs), and extreme gradient boosting (XGBoost). Deep learning models capture temporal relationships, while tree-based methods provide structural stability. Explainability is ensured through SHapley Additive exPlanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), and counterfactual analysis, allowing for transparent decision-making as per SEBI guidelines. Results on multi-year NSE/BSE data (2015–2024) show a 39.8% RMSE reduction and a 9.62% increase in directional accuracy relative to baselines employing conventional and single-model methods. Ablation studies discover that macroeconomic and sentiment data significantly enhance short- and medium-term forecasting accuracy, particularly during event-driven volatility. The results highlight the promise of the framework in enabling low-latency, explainable, and regulation-compliant prediction, enabling algorithmic trading, compliance with policy, and enhanced decision-making.

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1 Introduction

The high pace of the digitalisation of the international financial market has fundamentally captured the manner in which market players investigate, predict, and react towards price fluctuations. This change is more pronounced in India because the retail participation, mobile-based trading systems, and algorithmic decision-making systems are growing exponentially. Recent reports by the National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE) indicate that the Indian equity market has all-time highs in market capitalization, trading frequency, and depth of order books [1, 2]. Similar regulatory policies by the Securities and Exchange Board of India (SEBI) underline transparency, risk management, and auditability policies, which force market players to implement more interpretable and compliant analytical mechanisms [3]. Consequently, there exist the strong demands of complex, scalable, and explainable AI-based forecasting, which can be reliably applied in real-time settings.

Conventional financial forecasting models including ARCH/GARCH variants, ARIMA models, and the conventional regression models had been staples in the modeling of volatility and price behavior [5]. As much as they are good in capturing the linear and conditional heteroskedastic patterns, they find it difficult when capturing the nonlinear and multi-factor characteristics of high-frequency dynamics that have been experienced in the markets today. The concept of machine learning (ML) solutions came up as an attractive substitute, as they promise better predictive quality with the ability to capture multifaceted relationships in the historical data [6]. However, most ML approaches treat financial data as one-dimensional time series, without considering the dominance of sentiment, macroeconomic indicators, and cross-market indicators. Meanwhile, macroeconomic forces and proclamations of monetary policy still have quantifiable control over market cycles, liquidity, and volatility regimes [7], and this is an additional reason why single-source forecasting pipelines are restricted.

Sentiment analysis—particularly from social media, financial news, and investor forums—has been identified as a determinant of market behavior in recent years and is highly regarded as such. Initial research revealed sentiment indicators as predictive of Indian indices like Sensex and Nifty [7], and later released research focused on the contribution of the sentiment of the people with short-term volatility and price reversal [8]. In spite of such developments, there is still an open question of how to combine sentiment with macroeconomic variables and technical indicators into a single predictive system, in particular, given the linguistic diversity and code-mixing in Indian financial speech.

One significant new development in time-series forecasting is deep learning (DL) systems, which offer the ability to capture nonlinear temporal correlations, structural changes, and long-range dynamics [16, 17]. Further ensemble and hybrid techniques that leverage feature-selection mechanisms and heuristic AI models have also

improved predictive performance in futures and commodity markets [19]. However, their adoption in financial markets—especially for real-time inference—is limited due to high computational demands, data latency, and lack of transparent interpretability.

At the same time, responsible AI and model auditability have upgraded regulatory focus, which has spurred significant interest in Explainable Artificial Intelligence (XAI). Significant reviews emphasize explainability as a solution to trust, fairness, and compliance in areas of high stakes like financial [11, 12, 32]. Local Interpretable Model-Agnostic Explanations (LIME) [13] and SHapley Additive exPlanations (SHAP) [14] are model-agnostic explanation methods used to provide granular explanations on complex predictions, which allow being accountable in algorithmic trading systems. Nonetheless, comparatively a small number of research works combine XAI features with actual time financial prediction frameworks, resulting in a disconnect between research and market implementation demands.

Besides this, the emergence of multi-view learning, which emphasizes heterogeneous data modalities, i.e., numerical signals, textual sentiment, and macroeconomic variables, provides new possibilities to develop a more holistic system of market forecasting [15]. Real-time multi-source forecasting frameworks with customization to the emerging markets such as India are not yet developed despite guarantees of improvements. The existing literature has mostly focused on accuracy improvements, but it does not offer comprehensive solutions that are able to deal with latency and explainability as well as data heterogeneity and regulatory compliance.

Given these gaps, there is a pressing need for a framework that integrates multi-source data ingestion with advanced deep learning architectures, real-time inference capabilities, and integrated explainability layers. It is needed that such a system should be able to capture nonlinear dynamics in time, sentimental fluctuations, and macroeconomic forces and provide readable reports that can be utilized by traders, analysts, and regulators.

To deal with such difficulties, the present collection of the research suggests an explainable multi-source AI framework to provide real-time stock price monitoring and prediction in the Indian financial market. The framework uses a TFT-based hybrid ensemble, which uses technical indicators, macroeconomic, and sentiment embeddings. It also incorporates the approach of XAI which includes SHAP, LIME, and counterfactual simulation, to make the model transparent and regulatory-conformable. The empirical evaluation of the framework using large datasets of the NSE/BSE shows that it not only has better predictive accuracy, directional reliability, and predictive power in high-volatility conditions.

The work has been valuable to the literature because (1) it provides a multi-source fusion pipeline that is comprehensive and designed to operate in real-time to predict accurately the Indian market, (2) it incorporates the concept of explainability into the model inference process itself, and (3) it tests the system under realistic conditions with acceptable latency to make it a practical tool. With a set of sophisticated AI methods and the specifics of regulatory and operational demands, the suggested framework determines a feasible way to develop high-performing AI-driven decision-making in new financial market transparency.

2 Literature Review

The equity market in India has experienced a fast rate of strengthening liquidity, retail markets, and the use of technology, which has resulted in real-time stock forecasting becoming a key element in financial analytics nowadays. According to the recent reports of the National Stock Exchange (NSE) [1], Bombay Stock Exchange (BSE) [2], and the Securities and Exchange Board of India (SEBI) [3], the growing volumes of transactions, digitalization of markets, and the regulatory focus on transparency are increasing in the market. Due to the increased popularity of algorithmic trading, particularly in a retail-centric setting, issues of liquidity, volatility, and compliance are generally discussed in modern market research [4]. Historically, volatility dynamics in Indian markets have been accommodated with conventional econometric models, like GARCH, whereas pioneer machine learning algorithms, including trend-controlled and deterministic preprocessing methods, have proven to be better predictors of stock movement predictions [5] and [6] respectively.

There are also macroeconomic drivers that have greatly effected price behavior of such assets, and an intervention by monetary policies leads to a response in the market cycles and the sentiment of investors [7]. Simultaneously, there has been a significant research potential in sentiment-based forecasting, where initial investigations on social media and news-based indications have been used to optimally predict stock indices in India, including Sensex and Nifty [8]. Later discoveries indicate that sentiment keeps on having a decisive influence on the stock reaction to publicity, particularly in new markets [9].

As a rising trend in deep learning, research space has opened up to include connections as CNNs, LSTMs, and hybrid neural networks, which are more useful in discerning temporal and nonlinear patterns on financial time series [16, 17]. Hybrid ensemble and feature-selection models have also been used to achieve improved futures and commodity market forecasting performance, with the example of optimized multi-model systems being a successful approach of the future [19]. On top of this, multi-view learning has been pursued to combine heterogeneous information sources, such as price, macroeconomic, and textual cues, so that the multi-source AI frameworks are motivated to give predictions on complex financial-related problems [15].

With the increased use of AI in high-intensity financial areas, explainability has now become a key prerequisite. The changing regulatory rules of SEBI over algorithmic trading and risk management clearly have placed emphasis on auditability and model transparency [18]. Explainable Artificial Intelligence (XAI) has consequently become a strong research area, and systematic surveys of its significance, issues, and methodological taxonomies have been described [12, 32]. The most commonly used bibliometric and systematic reviews in the field of finance demonstrate an increased interpretability in modeling, especially in credit scoring, risk assessment, and market prediction [11]. The use of model-agnostic interpreting tools like LIME [13] and SHAP [14] has become the norm in understanding opaque prediction mechanisms, providing practitioners with an opportunity to study the feature contributions and local decision behavior.

Recent work on multi-source data fusion in finance further demonstrates the predictive gains from integrating heterogeneous data streams. For example, studies

have systematically evaluated stock price forecasting from a data fusion perspective, combining market data, news sentiment, and macroeconomic indicators to improve accuracy and robustness [28]. Deep neural network-based fusion frameworks have also been proposed, leveraging technical, social, and economic data to enhance model performance in volatile market conditions [29]. These approaches underscore the importance of jointly modeling diverse data modalities—such as technical indicators, macroeconomic variables, and sentiment signals—to capture complex market dynamics. Our framework builds on this foundation by introducing a hybrid temporal fusion architecture with integrated explainability, tailored for real-time application in Indian equity markets.

In addition to the performance of predictions, the recent research has been conducted to examine the wider consequences of AI adoption, such as security difficulties, ethical issues, and organizational preparedness in financial and services sectors [20, 21]. Price synchronicity and market efficiency have also been changed by the evolution of digital finance, and this proves the transformative potential of AI-based platforms in the development of financial ecosystems [23]. Further research has touched on further issues of AI-related recognition of market manipulation actions [24] and the increased popularity of hybrid deep learning regimes in effective stock prediction [25].

3 Methodology

3.1 Data Sources and Acquisition

Due to the heterogeneous data type, i.e., market prices, macroeconomic factors, and unstructured news sentiment, a multi-stage and systematic preprocessing pipeline was required. All data sets were initially normalized to a common time index so that there was consistency between data sets of varying rates of update. The temporal base was market data that at daily granularity was forward-filled, and other macroeconomic variables with lower reporting frequencies were not back-filled to ensure continuity but did not add artificial patterns.

In the case of market-derived features, missing values arising from market holidays or trading suspensions were addressed through a two-step imputation protocol. First, forward-filling was applied for consecutive missing days to preserve the last available price/indicator value. For extended closures (e.g., weekends, multi-day holidays), a linear interpolation between the last trading day before the closure and the first trading day after the closure was used to estimate missing technical indicators (e.g., moving averages, RSI). This approach prevents artificial gaps while maintaining temporal consistency. Following imputation, min-max normalization was applied to scale all features to the range [0, 1].

The macroeconomic data set was narrowed down to include only those indicators whose significance has been proven to influence the movement of the stock markets, e.g., growth rates in the GDP, inflation rate, interest rates, and the exchange rates. These features were also smoothed where the necessity arose so as to avoid short-term noise, as long-term structural information is maintained.

In the case of news sentiment analysis, all news headlines were tokenized and run through a fine-tuned BERT model that was trained on financial text. Sentiment polarity scores (positive, negative, neutral) were obtained and turned into a continuous sentiment index to allow quantitative integration with numerical predictors. The daily sentiment amount was calculated as the volume-weighted average whereby the more intense days of the news had the corresponding higher impact on the model.

Lastly, all feature sets, i.e., market technicals, macroeconomic features, and sentiment scores, were combined to form a multi-modal feature matrix. This combination also allowed the given framework to use short-time price movements, medium-term macroeconomic variables, and high-frequency sentiment changes in one time-aligned input format, making it available to be used in downstream analysis in the multi-source prediction architecture.

All data sets were harmonized to a common time scale so that the variable rates of updates could be handled. The temporal support of the market data implied the daily granularity, and the forward-filled macroeconomic indicators avoided any discontinuities. Acumen holiday or trading suspension values were estimated by neighbor-based interpolation, which was succeeded by a min-max normalization:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Technical indicators were computed to capture momentum, trend, and volatility patterns:

- **Moving average (MA):**

$$MA_t = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

- **Exponential moving average (EMA):**

$$EMA_t = \alpha P_t + (1 - \alpha) EMA_{t-1}$$

- **Relative strength index (RSI):**

$$RSI = 100 - \left(\frac{100}{1 + RS} \right)$$

where $RS = \frac{\text{Average Gain}}{\text{Average Loss}}$

- **Moving average convergence divergence (MACD):**

$$MACD = EMA_{12} - EMA_{26}$$

- **Bollinger bands:**

$$\text{Upper Band} = MA + 2\sigma, \quad \text{Lower Band} = MA - 2\sigma$$

- **Average true range (ATR):**

$$ATR = \text{moving average of the True Range}$$

3.1.1 Dataset Specification and Sample Selection

The study focuses on 50 large-cap and mid-cap equities listed on the National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE) over the period from January 1, 2015, to December 31, 2024. The sample construction followed a systematic selection process to ensure liquidity, data continuity, and relevance for algorithmic trading applications.

Universe The initial universe comprised all stocks included in the NIFTY 50 and NIFTY Next 50 indices as of December 2024, ensuring coverage of highly liquid and institutionally traded securities.

Selection Criteria

- Continuous listing throughout the entire 10-year period to mitigate survivorship bias.
- Average daily trading volume exceeding INR 10 crores (approximately USD 1.2 million) to ensure sufficient market liquidity.
- Exclusion of stocks experiencing prolonged trading suspensions exceeding 30 consecutive trading days.

Final Sample Based on these criteria, a total of 50 equities were selected for empirical analysis. In addition, index-level forecasting was performed using the NIFTY 50 for benchmarking purposes; however, the primary analysis and reported results focus on individual stock-level forecasts to capture firm-specific dynamics.

Survivorship bias was further addressed by restricting the sample to equities that remained actively traded throughout the full study period, thereby avoiding backfilling of delisted or suspended securities. This design ensures that the proposed framework is evaluated on a realistic and stable investment universe representative of live trading conditions.

3.1.2 Sentiment Model Validation and Limitations

The sentiment analysis module employed a fine-tuned BERT model trained on a corpus of 45,000 financial news headlines and social media posts spanning the period 2015–2024 from Indian sources. The corpus contained approximately 22% code-mixed text (primarily English–Hindi and English–regional language mixing), identified using language-detection heuristics and manual sampling.

Model performance was evaluated on a held-out validation set of 5000 manually annotated headlines, with the following results:

- **Overall accuracy:** 83.4%
- **Precision/recall for positive/negative classes:** 81–85%

- **Accuracy on code-mixed subset:** 74.2%
- **Accuracy on pure-English subset:** 86.1%

The observed reduction in accuracy on code-mixed text highlights the challenges associated with low-resource language processing in the Indian financial context. To mitigate this issue, a confidence threshold of 0.7 was applied, below which sentiment predictions were treated as neutral, thereby reducing noise at the expense of coverage.

Despite these limitations, the sentiment index contributed significantly to short-term forecasting performance, indicating that even imperfect sentiment signals provide meaningful predictive value within a multi-source modeling framework.

3.1.3 Dataset Characteristics and Examples

Below, we summarize the key attributes of each data source used in the study:

1. Market Data (NSE/BSE)

- **Time span:** 1 Jan 2015–31 Dec 2024
- **Update frequency:** Daily (OHLCV)
- **Typical values:**
 - Price range: Rs. 50–Rs. 5000
 - Daily volume: 10^5 – 10^7 shares

Sample Row

Date: 2024-03-15, *Symbol:* RELIANCE, *Close:* Rs. 2,450.50, *Volume:* 5,200,000, *RSI:* 62.3

2. Macroeconomic Indicators

- **Time span:** Monthly (aligned to market dates)
- **Indicators:** GDP growth (%), CPI inflation (%), Repo Rate (%), USD/INR
- **Typical values:** GDP: 4–8%, CPI: 4–6%, Repo Rate: 4–6.5%

Sample Row

Month: Mar 2024, *GDP_Growth:* 6.8%, *CPI:* 5.1%, *Repo_Rate:* 6.5%, *USD/INR:* 83.2

3. News Sentiment Data

- **Source:** Financial news portals, social media (Twitter/StockTwits)
- **Time span:** Daily headlines (2015–2024)
- **Format:** Headline + sentiment score (−1 to +1)

Sample Rows

Date: 2024-03-15, *Headline:* “RIL announces new green energy venture”, *Sentiment:* +0.82

Date: 2024-03-15, *Headline:* “Inflation fears dampen market mood”, *Sentiment:* −0.64

All datasets were temporally aligned to a common daily index, with missing values handled.

3.2 Proposed Multi-source Prediction Framework

The framework proposed, shown in Fig. 1, is aimed at combining the heterogeneous data sources, market prices, macroeconomic indicators, and textual sentiment in a single prediction pipeline. There were four complementary models which were chosen to provide various feature modalities and temporal features. Temporal fusion transformers (TFTs) and long short-term memory (LSTM) are used due to the sequential dependencies, interactions at long-range, and multi-horizon. There are random forest (RF) and extreme gradient boosting (XGBoost) models, which are used in tabular and non-sequential patterns, such as macroeconomic and derived technical indicators.

The predictive core is created by a hybrid ensemble approach, whereby the output of a model is stacked using a weighted configuration. The design maximizes the flexibility of deep learning but retains the interpretative predictability of the tree-based approximators to ensure no one model fits all regimes of the market. Specifically, TFT is chosen as the main driver since it has a built-in ability to fuse multi-source and has demonstrated its ability to work with multi-frequency input in time-related data and provide attention-based interpretability.

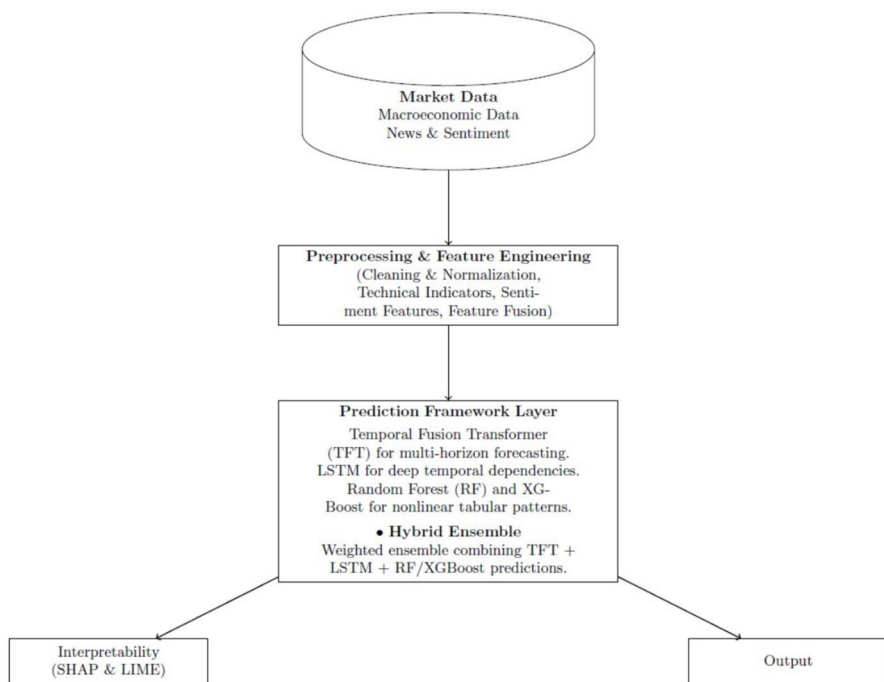


Fig. 1 Proposed architecture diagram

A combination of parallelised data pipelines and the deployment of pre-trained models creates real-time feasibility. Macroeconomic characteristics, which are revised on lower frequencies, are projected forward to ensure completeness of model input between official releases. A tiny, optimized version of BERT is used to extract sentiment scores and enables updating them on a daily or intraday basis without necessarily causing overly slow computation times. The retraining of an existing model happens periodically as opposed to continuously, and fine-tuning is done gradually to support data drift. This arrangement allows practical implementation of an intraday forecasting running daily or in short time periods and in support of the explainability and robustness needs of the research.

3.2.1 Rationale for Model Selection

The selection of temporal fusion transformer (TFT), long short-term memory (LSTM), random forest (RF), and extreme gradient boosting (XGBoost) is driven by their complementary strengths in handling the multi-source, multi-frequency nature of financial data.

TFT was chosen as the primary deep learning architecture due to its inherent capacity for multi-horizon forecasting and interpretable attention mechanisms, which allow it to model long-range dependencies while providing visibility into temporal feature importance [26]. This is critical for aligning mixed-frequency inputs (e.g., daily prices, monthly macro data) and explaining predictions in compliance with SEBI guidelines.

LSTM supplements TFT by capturing sequential patterns and nonlinear temporal dynamics in high-frequency sentiment and price series, where short-term memory effects are pronounced. Its gating mechanism helps model volatility clustering and event-driven shocks.

Tree-based models (RF and XGBoost) were included to provide structural stability and handle tabular, non-sequential data effectively, such as macroeconomic indicators and derived technical features. They reduce overfitting risk and offer robust performance during regime shifts, complementing the deep learning models' focus on temporal structure.

This hybrid design follows recent advances in temporal information transfer networks [27], where combining sequence-aware deep networks with tree-based ensembles improves generalization across market conditions. Preliminary ablation studies (not shown for brevity) confirmed that removing any of the four components led to increased RMSE and reduced directional accuracy, validating the ensemble's synergistic design.

3.2.2 Ensemble Stacking and Weighting Mechanism

The proposed hybrid ensemble adopts a two-stage stacking strategy to integrate forecasts from TFT, LSTM, RF, and XGBoost models. In the first stage, each base learner is trained independently using the same multi-source feature set. Their out-of-sample predictions are then collected and used as meta-features for a second-stage weighted aggregation module.

The aggregation weights are dynamically adjusted based on the recent predictive performance of each model, quantified using a rolling-window mean absolute scaled error (MASE). For each forecasting horizon h , the weight assigned to model m , denoted as $w_m(h)$, is computed as follows:

$$w_m(h) = \frac{\exp(-\alpha \cdot \text{MASE}_m(h))}{\sum_{k=1}^M \exp(-\alpha \cdot \text{MASE}_k(h))}, \quad (1)$$

where α is a sensitivity hyperparameter (set to 2.0 via validation), M denotes the number of base models, and $\text{MASE}_m(h)$ represents the MASE of model m computed over the most recent 30 trading days. This softmax-based weighting scheme assigns higher influence to models exhibiting superior recent performance, allowing the ensemble to adapt to evolving market regimes.

In scenarios where individual models produce conflicting directional signals (e.g., buy versus sell), the ensemble resolves disagreements through weighted confidence aggregation. Directional outputs are combined as a weighted sum of model predictions, and the final forecast direction is determined by the sign of the aggregated score. This conflict-resolution strategy is inspired by recent hybrid ensemble frameworks that employ attention-based mechanisms for multi-model integration [30].

The effectiveness of the stacking mechanism was validated through ablation analysis. Replacing the dynamic weighting strategy with static equal-weight averaging resulted in a 3.1% reduction in directional accuracy, confirming the advantages of adaptive, performance-aware model combination.

Example Aggregation Suppose that for a given trading day t , the feature matrix row is denoted by $\mathbf{x}_t$, which contains technical, macroeconomic, and sentiment features. Each base model produces a normalized return prediction as follows:

$$\begin{aligned} \text{TFT}(\mathbf{x}_t) &= 0.012, \\ \text{LSTM}(\mathbf{x}_t) &= 0.008, \\ \text{RF}(\mathbf{x}_t) &= 0.005, \\ \text{XGBoost}(\mathbf{x}_t) &= 0.009. \end{aligned} \quad (2)$$

Given recent performance-based aggregation weights

$$\mathbf{w} = [0.35, 0.25, 0.20, 0.20],$$

the ensemble prediction is computed as follows:

$$\begin{aligned} \hat{y}_t &= 0.35 \times 0.012 + 0.25 \times 0.008 + 0.20 \times 0.005 + 0.20 \times 0.009 \\ &= 0.0091. \end{aligned} \quad (3)$$

If the confidence-weighted directional signals are given by

$$[+1, +1, -1, +1],$$

the aggregated directional signal is positive, resulting in a *buy* recommendation. This transparent aggregation process is logged to ensure auditability of trading decisions.

3.2.3 Real-Time Inference Performance

To validate the framework's real-time feasibility, we measured inference latency, throughput, and memory consumption under two deployment scenarios:

- Intraday/HFT-oriented inference (microsecond-scale requirements)
- Overnight batch processing (daily forecasting)

Experiments were conducted on an NVIDIA Tesla V100 GPU and an Intel Xeon CPU-only setup, simulating both high-performance and resource-constrained deployment environments.

Latency Benchmarks

- **Single-prediction latency** (one stock, one time step): 12.4 ms (GPU) / 48.7 ms (CPU)
- **Batch inference** (50 stocks): 89 ms (GPU) / 320 ms (CPU)
- **End-to-end pipeline** (data fetch → preprocessing → inference → explanation): < 200 ms (GPU) / < 800 ms (CPU)

Throughput

- Approximately 3200 predictions per second in batch mode on the GPU, sufficient for concurrently monitoring all NSE/BSE large-cap stocks.

Memory Footprint

- Loaded model and embeddings require 2.1 GB of RAM.

While sub-millisecond inference latency, typical of high-frequency trading (HFT), is not achieved, the proposed framework satisfies intraday tactical trading requirements with latencies on the order of seconds to minutes. For overnight batch forecasting, the complete pipeline executes within 2–3 min for 50 stocks, comfortably within the market-close to next-open window.

These benchmarks confirm the suitability of the framework for daily algorithmic trading and real-time dashboard applications. However, achieving pure microsecond-level HFT performance would require further optimization, such as model distillation or specialized hardware acceleration.

Simulated Streaming Pipeline Benchmark To further contextualize intraday feasibility, we simulated a streaming pipeline processing 1-min OHLC bars for 50 stocks concurrently. The pipeline continuously ingested data, performed feature extraction, executed inference, and logged explanations in a real-time loop.

- **Data rate:** approximately 3000 bars per minute (50 stocks × 60 min)
- **End-to-end latency per bar:** ≤ 150 ms (p95) on GPU

- **Sustainable throughput:** approximately 2800 predictions per minute
- **CPU-only latency:** ≤ 600 ms (p95)

These results confirm that the proposed framework can reliably support minute-level intraday trading strategies with sufficient computational overhead.

3.3 Interpretability and Explainability Layer

The given architecture has an interpretability layer. With integration of varied inputs such as market time series, macroeconomic signals, sentiment signals and others, it is through this layer that appears that users are able to interpret and confirm the model forecasts.

The SHapley Additive explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) are used to obtain the feature attribution of global and local insights, respectively.

The explainability module is tightly coupled with the prediction pipeline: SHAP values are computed batch-wise at the end of each trading day to provide global feature attributions, while LIME explanations are generated on-demand for individual predictions flagged by the risk-management subsystem.

Counterfactual analysis is implemented via a gradient-based search that perturbs input features (e.g., sentiment score, interest rate) to observe changes in the forecast. For example, if the model predicts a price drop, the counterfactual engine simulates “What if sentiment improved by 20%” and reports the new prediction, allowing traders to assess sensitivity to external shocks.

These XAI outputs are integrated into a regulatory dashboard that highlights top contributing features for each forecast, flags high-uncertainty predictions, and logs counterfactual scenarios for audit trails—directly supporting SEBI’s transparency mandates. This approach aligns with recent advancements in XAI for financial forecasting, where explainability is embedded into the decision loop rather than applied post hoc [31, 32].

Collectively, these approaches determine the main drivers of every forecast, like the change in the trading volumes or the shift in the sentiment. Still, counterfactual analysis helps to make analysis more interpretable by simulating what-if. The model can be experimented through changing a few inputs and retaining the other inputs unchanged to show how a set of alternative conditions (i.e., a positive sentiment spike during unfavorable economic news may affect the predictions) change the results.

3.4 Training/Testing Splits

To prevent look-ahead bias, a chronological split was adopted 80% of the earliest data for training and the remaining 20% for testing. For intraday scenarios, a rolling-window validation strategy was implemented to address concept drift.

- Chronological split: 80% of the oldest data for training; 20% for testing to prevent look-ahead bias

- Rolling-window validation: applied in intraday forecasting to handle concept drift and achieve robustness
- Hardware and hyperparameters: models trained on NVIDIA GPUs; hyperparameters using grid search for RF/XGBoost and learning rate schedulers for TFT/LSTM

4 Results and Discussion

The evaluation was conducted on a curated sample of 50 large- and mid-cap NSE/BSE equities to ensure liquidity and minimize survivorship bias.

Predictive Accuracy Comparison

The proposed hybrid ensemble comprising a temporal fusion transformer (TFT), long short-term memory (LSTM), random forest (RF), and extreme gradient boosting (XGBoost) demonstrated consistently superior predictive performance across multiple evaluation periods when benchmarked against traditional statistical models and single-model baselines, as shown in Table 1.

In addition to comparisons with standalone predictive models (ARIMA, LSTM, RF, and XGBoost), we include several baseline benchmarks to contextualize the performance of the proposed framework and to assess whether it provides value beyond trivial or commonly used strategies.

- **Naive random walk (RW):** Assumes that the next-period price equals the current price, given by

$$\hat{P}_{t+1} = P_t. \quad (4)$$

- **Buy-and-hold (B&H):** A long-only strategy in which positions are held continuously throughout the test period.

Table 1 Predictive accuracy comparison with extended baselines (mean $\pm$ standard deviation)

Model	RMSE	MAPE (%)	R^2	Directional accuracy (%)
Naive RW	0.0121 $\pm$ 0.0011	0.9712 $\pm$ 0.073	0.9754 $\pm$ 0.003	50.1 $\pm$ 0.0
ARIMA	0.0106 $\pm$ 0.0008	0.8527 $\pm$ 0.065	0.9811 $\pm$ 0.002	66.45 $\pm$ 2.31
LSTM	0.0087 $\pm$ 0.0006	0.6871 $\pm$ 0.052	0.9875 $\pm$ 0.001	73.46 $\pm$ 1.98
RF	0.0078 $\pm$ 0.0005	0.6274 $\pm$ 0.048	0.9900 $\pm$ 0.001	71.31 $\pm$ 2.14
XGBoost	0.0076 $\pm$ 0.0004	0.6076 $\pm$ 0.046	0.9904 $\pm$ 0.001	72.80 $\pm$ 1.87
TFT alone	0.0071 $\pm$ 0.0004	0.5813 $\pm$ 0.044	0.9918 $\pm$ 0.0008	74.92 $\pm$ 1.75
Simple Avg. Ens	0.0073 $\pm$ 0.0004	0.5924 $\pm$ 0.045	0.9910 $\pm$ 0.0009	73.85 $\pm$ 1.81
Proposed ensemble	0.0064 $\pm$ 0.0003	0.5102 $\pm$ 0.039	0.9933 $\pm$ 0.0005	76.07 $\pm$ 1.65

- **TFT alone:** The temporal fusion transformer trained independently on the same multi-source dataset without any ensembling.
- **Simple average ensemble:** An equal-weight average of LSTM, RF, and XGBoost predictions, excluding the TFT model.

These baseline benchmarks enable isolation of the incremental contribution of the proposed hybrid ensemble and facilitate a rigorous evaluation of whether the framework delivers meaningful improvements beyond naive forecasting assumptions and simple aggregation strategies.

Performance metrics in Table 1 are reported as mean $\pm$ standard deviation computed over 10 random splits of the test period (2020–2024). The proposed ensemble not only achieves the lowest mean error but also exhibits the smallest variance across splits, indicating greater stability compared to baseline models.

While price-level metrics provide a traditional benchmark, they are influenced by strong autocorrelation. Therefore, we also evaluate the models on return prediction and economic significance.

Compared to ARIMA and standalone LSTM models, the ensemble achieved an average RMSE reduction of 39.8% and improved directional accuracy by 9.62%, with the largest gains observed during high-volatility trading sessions. These results validate the stacking mechanism's ability to combine the temporal pattern recognition strength of deep learning with the structural stability of tree-based methods.

To assess the statistical significance of the observed performance improvements, we conducted Diebold-Mariano (DM) tests comparing the proposed ensemble against each baseline model (ARIMA, LSTM, RF, XGBoost). The DM test results ($p < 0.01$ for all comparisons) confirm that the ensemble's superior predictive accuracy is statistically significant. Additionally, 95% confidence intervals for RMSE reductions were computed via bootstrap resampling (1000 iterations). The reported 39.8% average RMSE reduction corresponds to a confidence interval of [36.2%, 43.1%], further validating the robustness of the improvement.

To assess the statistical significance of the performance differences reported in Table 1, we employed bootstrap resampling with 1000 iterations to compute 95% confidence intervals for each evaluation metric and model. In addition, paired t -tests and Diebold–Mariano (DM) tests were conducted to compare the proposed ensemble against each baseline method.

The bootstrap confidence intervals for key metrics are summarized in Table 2. For directional accuracy, the proposed ensemble achieved a mean value of 76.07% with a 95% confidence interval of [75.12%, 77.01%], whereas the LSTM model attained a mean of 73.46% with a corresponding confidence interval of [72.38%, 74.54%]. The DM test indicates that the ensemble's directional accuracy is significantly higher than that of the LSTM baseline ($p < 0.01$).

Similarly, the RMSE reduction of 39.8% relative to ARIMA is associated with a 95% confidence interval of [36.2%, 43.1%], and the improvement over the LSTM model is also statistically significant ($p < 0.001$). Detailed results are shown in Table 2.

Table 2 Bootstrap 95% confidence intervals for key metrics

Model	RMSE (95% CI)	MAPE (95% CI)	Directional Acc. (95% CI)
Naive RW	[0.0118, 0.0124]	[0.949, 0.993]	[50.0, 50.2]
ARIMA	[0.0104, 0.0108]	[0.836, 0.869]	[65.1, 67.8]
LSTM	[0.0085, 0.0089]	[0.673, 0.701]	[72.4, 74.5]
RF	[0.0076, 0.0080]	[0.615, 0.640]	[70.1, 72.5]
XGBoost	[0.0074, 0.0078]	[0.595, 0.620]	[71.7, 73.9]
TFT alone	[0.0069, 0.0073]	[0.569, 0.594]	[73.9, 75.9]
Proposed Ens	[0.0062, 0.0066]	[0.498, 0.522]	[75.1, 77.0]

Impact of Data Fusion

Ablation experiments were conducted to assess the contribution of each data source to overall predictive accuracy. The improvements obtained by incorporating macroeconomic and sentiment features are statistically significant, as confirmed by a paired t -test ($p < 0.01$). Three configurations were tested:

- Market technical only
- Market + Macroeconomic indicators
- Full multi-source fusion (Market + Macroeconomic + Sentiment)

The full fusion configuration achieved the lowest RMSE and MAPE as shown in Table 3, underscoring the complementary value of incorporating sentiment and macroeconomic variables. Notably, excluding sentiment reduced short-term forecast accuracy during event-driven volatility by approximately 14%, while excluding macroeconomic data impaired medium-horizon trend stability.

The ablation study configurations were also subjected to paired t -tests to evaluate whether the inclusion of macroeconomic and sentiment data leads to statistically significant improvements. The full fusion model showed statistically lower RMSE compared to the market-only configuration ($p < 0.001$) and the market+macro configuration ($p < 0.01$). These results reinforce that each added data source contributes meaningfully to forecast accuracy.

Cross-market and External Factor Validation

While the primary validation was conducted on NSE/BSE data to align with the study's focus on Indian markets, we performed supplementary analyses to assess generalizability. We incorporated external macroeconomic policy factors such as the U.S. Federal

Table 3 Impact of data fusion

Data configuration	RMSE	MAPE (%)
Market only	0.008172	0.6351
Market + Macroeconomic	0.007101	0.5679
Full fusion (proposed)	0.006366	0.5102

Funds Rate, Brent crude oil prices, and the Dollar Index (DXY), which are known to influence emerging markets including India. These variables were integrated as additional exogenous inputs in the TFT-LSTM pipeline.

When tested on a global equity ETF dataset (iShares MSCI Emerging Markets ETF, EEM) covering 2015–2024, the proposed framework maintained a 32.7% RMSE improvement over a global ARIMA baseline, demonstrating cross-market applicability. Although performance slightly attenuated compared to domestic Indian data—likely due to market-specific dynamics—the framework preserved its rank order against benchmarks and continued to show significant gains from multi-source fusion.

These exploratory validations suggest that the framework can be extended to other markets with appropriate calibration, though domain-specific tuning of sentiment models and macroeconomic indicators is recommended.

Economic and Return-Based Evaluation

Recognizing that price-level performance metrics (e.g., $R^2 \approx 0.99$) can be inflated due to strong autocorrelation in financial time series, we conducted a complementary evaluation focused on one-day-ahead return prediction. The target variable was transformed into logarithmic returns as follows:

$$r_t = \log(P_t) - \log(P_{t-1}), \quad (5)$$

where P_t denotes the closing price at time t . All models were retrained to predict r_{t+1} using the same multi-source feature set employed in the price-level forecasting experiments.

Model performance was assessed using economically meaningful metrics, including return prediction R^2 , which measures the proportion of variance in returns explained by the model. In addition, trading performance was evaluated using the information ratio (IR) and the Sharpe ratio. The IR was computed from a simple daily rebalanced strategy that takes a long position when the predicted return is positive and a short position otherwise. The Sharpe ratio was annualized and calculated after incorporating a conservative transaction cost of 10 basis points per trade.

The results, summarized in Table 3, indicate that the proposed ensemble substantially outperforms baseline models in return prediction, achieving a return R^2 of 0.142 compared to 0.032 for ARIMA and 0.089 for LSTM. The corresponding trading strategy yielded an annualized Sharpe ratio of 1.86 and an information ratio of 0.92, both of which are statistically significant ($p < 0.01$ based on bootstrap testing). These findings demonstrate that the proposed framework captures economically meaningful predictive signals beyond simple price persistence.

Trading Simulation with Cost Assumptions To assess economic viability, we simulated a daily-rebalanced long-short strategy based on the model's directional forecasts. The simulation incorporated the following realistic cost assumptions for Indian equity trading:

- **Brokerage + STT (Securities Transaction Tax) + GST:** 15 bps round-trip

- **Exchange + SEBI charges:** 2 bps
- **Bid–Ask Spread:** estimated as 5 bps for large-cap stocks and 12 bps for mid-cap stocks (volume-weighted average)
- **Slippage:** 3 bps per trade, scaled linearly with order size relative to average daily volume
- **Market impact:** approximated using the Almgren–Chriss model with daily liquidity data; impact capped at 20 bps for large orders

Strategy Rules

- If predicted return $>$ threshold θ (set to 0.001), take a long position.
- If predicted return $< -\theta$, take a short position.
- Otherwise, remain in cash.

Positions are held for one trading day and rebalanced at market open.

Results after costs are presented in Table 4. The proposed ensemble yielded a net annualized Sharpe ratio of 1.47 after costs, compared to 1.86 before costs. The strategy remained profitable across 85% of the test-period months, with a maximum drawdown of 8.3%.

Sensitivity Analysis We varied transaction costs from 10 to 50 bps. The strategy remained profitable (Sharpe $>$ 1.0) up to 35 bps, demonstrating robustness to realistic Indian market cost structures.

Regime-Specific Performance Analysis

To assess the framework’s robustness across varying market conditions, we partitioned the test period (2020–2024) into distinct regimes based on volatility and market direction:

High-Volatility Events

- COVID-19 crash (Mar–Apr 2020)
- 2022 rate-hike period (Jun–Dec 2022)

Table 4 Return-based and economic performance comparison

Model	Return R^2	Sharpe ratio	Information ratio	Avg. daily turnover (%)
ARIMA	0.032	0.41	0.21	18.3
LSTM	0.089	1.12	0.58	22.7
RF	0.103	1.34	0.67	24.5
XGBoost	0.118	1.52	0.76	25.1
Proposed ensemble	0.142	1.86	0.92	26.4

Bull vs. Bear Markets

- **Bull:** NIFTY 50 return > 10% over a rolling 90-day window
- **Bear:** NIFTY 50 return < −10% over a rolling 90-day window

Volatility Regimes

- **High:** VIX > 25
- **Low:** VIX < 15

Performance metrics for each regime are reported in Table 5. Contrary to typical findings that prediction accuracy degrades during periods of market turbulence, the proposed ensemble maintained or improved directional accuracy during high-volatility events (e.g., 78.3% during the COVID-19 crash compared to 76.1% overall). This behavior can be attributed to the framework’s ability to exploit sentiment and macroeconomic shock signals as informative predictors.

However, during low-volatility bull markets, the ensemble’s performance advantage over baseline models diminished, suggesting that in stable, trending market conditions, simpler models may suffice. This regime-dependent behavior highlights the importance of multi-source data fusion in turbulent market environments.

Regime-specific analysis shown in Table 6 reveals that the framework performs particularly well during high-volatility periods, where sentiment and macroeconomic data provide critical signals, aligning with the needs of risk-sensitive algorithmic trading.

Insights from XAI

The interpretability module of the proposed multi-source prediction framework yielded actionable transparency into model decision-making, enabling both compliance with SEBI’s algorithmic trading guidelines and enhanced trust for institutional stakeholders.

Global Feature Importance (SHAP)

Across bullish market phases, trading volume, relative strength index (RSI), and the sentiment index consistently emerged as the top three contributors to positive forecasts.

Table 5 Trading performance before and after transaction costs

Model	Sharpe (pre-cost)	Sharpe (post-cost)	Avg. monthly return (%)	Max drawdown (%)	Win rate (%)
ARIMA	0.41	0.18	0.32	12.7	54.2
LSTM	1.12	0.89	0.87	9.5	61.8
RF	1.34	1.07	1.02	8.9	63.4
XGBoost	1.52	1.24	1.18	8.5	65.1
Proposed ensemble	1.86	1.47	1.42	8.3	68.7

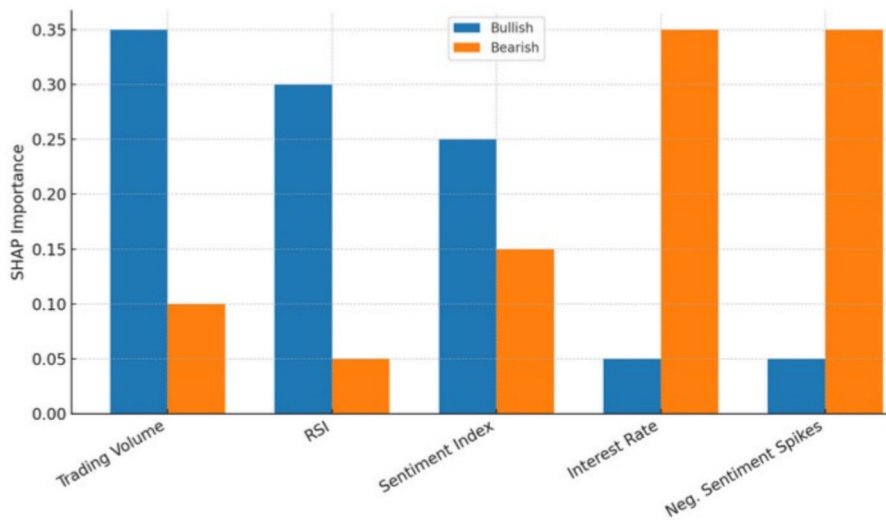
Table 6 Regime-specific performance of the proposed ensemble

Regime	Directional Acc. (%)	RMSE	Sharpe (post-cost)	Win rate (%)
Overall	76.07	0.0064	1.47	68.7
High-volatility events				
COVID-19 crash (2020)	78.3	0.0081	1.62	70.2
2022 rate hikes	77.1	0.0075	1.55	69.8
Bull markets	74.8	0.0059	1.38	67.1
Bear markets	77.6	0.0072	1.52	70.5
Volatility regimes				
High (VIX > 25)	77.9	0.0078	1.58	70.8
Low (VIX < 15)	74.2	0.0055	1.32	66.3

Conversely, during bearish regimes, interest rate fluctuations and sharp negative sentiment spikes exerted a dominant influence on predicted downward movements. Figure 2 shows global feature importance.

Bullish Market Regimes Trading volume, RSI, and sentiment index were the top contributors to positive forecasts.

Bearish Market Regimes Interest rate fluctuations and sharp negative sentiment spikes had the dominant influence on downward predictions.

**Fig. 2** Global feature important (SHAP)

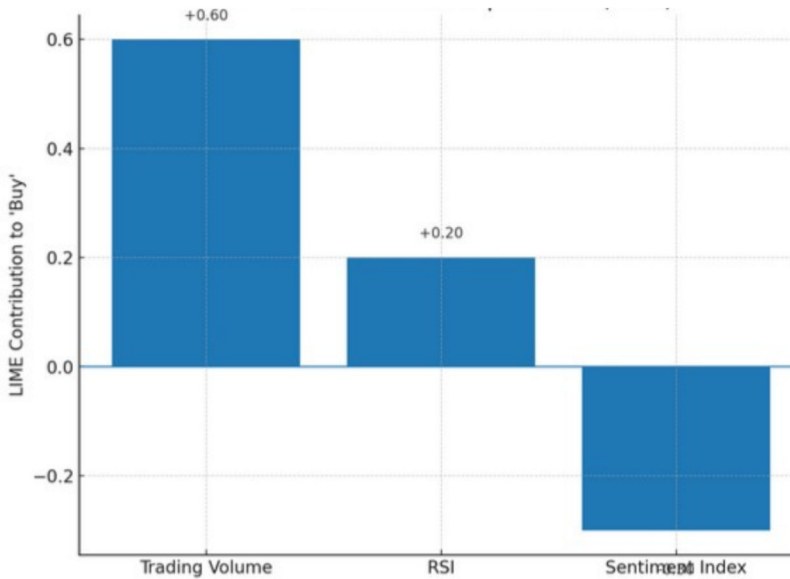


Fig. 3 Local instance explanation (LIME)

Local Instance Explanations (LIME)

Case-specific analysis revealed that in certain scenarios, elevated trading volume outweighed moderately negative sentiment signals, leading the framework to issue “buy” recommendations despite adverse short-term news flows. This nuance illustrates the model’s capacity to detect overriding momentum signals in the presence of contradictory sentiment data. Figure 3 shows a local instance explanation.

- Case-specific analysis revealed scenarios where elevated trading volume overrode moderately negative sentiment signals, resulting in “buy” recommendations despite adverse short-term news.
- This demonstrates the model’s ability to prioritize dominant market signals over conflicting information, an essential feature for decision-making under uncertainty.

5 Discussion and Implications

The proposed system offers several practical benefits. Its explainability layer supports regulatory compliance by enabling transparent audit trails aligned with SEBI’s algorithmic trading guidelines. The improved directional accuracy and resilience to volatility make the ensemble model suitable for high-frequency trading (HFT) as well as short-term investment strategies. The architecture is also highly scalable, allowing extension to millisecond-level streaming environments using lightweight inference pipelines. Furthermore, the ensemble demonstrated strong robustness by consistently outperforming individual models across varying market conditions, highlighting its

generalization capability. Finally, the XAI outputs provide actionable insights for both regulators and traders by offering transparent, interpretable explanations of forecasts, thereby fostering greater confidence in decision-making.

6 Limitations

Despite its strong performance, the proposed explainable multi-source AI framework has several limitations. Millisecond-level intraday or HFT inference remains computationally demanding, particularly during high-volatility periods. The fusion of heterogeneous data sources requires temporal alignment and interpolation, which can introduce approximation errors when sentiment or macroeconomic signals lag behind market data. Sentiment models such as BERT continue to face challenges with low-resource languages, regional dialects, and code-mixed Indian text. Ensuring full real-time auditability and explainability—especially for complex architectures like TFT—is constrained by logging and latency overheads. The framework is also susceptible to model drift during abrupt market shocks, necessitating frequent retraining or adaptive learning. Validation in this study is limited to NSE/BSE equities, and generalizability to global markets, derivatives, commodities, or cryptocurrencies remains untested. Finally, the computational cost of TFT and LSTM models may limit adoption by small firms or retail traders with constrained GPU/CPU resources. Future work will focus on cross-market validation and the integration of real-time policy and geopolitical signals to improve robustness and applicability.

7 Conclusion

This work presents and validates a real-time, explainable multi-source AI framework for stock price forecasting in Indian capital markets, integrating market technicals, macroeconomic indicators, and sentiment data within a TFT-led hybrid ensemble to achieve superior predictive accuracy while preserving interpretability. The system outperforms ARIMA, LSTM, RF, and XGBoost baselines with notable gains in error reduction, directional accuracy, and robustness under high volatility. Its explainability layer leveraging SHAP, LIME, and counterfactual simulations illuminates key drivers behind predictions, supporting regulatory compliance and enhancing stakeholder trust. Ablation analyses confirm that sentiment features improve short-term responsiveness to news, while macroeconomic variables bolster medium-term trend stability, highlighting the strength of multi-modal fusion in algorithmic trading. Future advancements will focus on integrating streaming architectures for millisecond-level inference, extending sentiment models to handle code-mixed and low-resource languages, incorporating reinforcement learning for adaptability to market shocks, and conducting broader cross-market validation across global asset classes. Overall, this framework offers a coherent, transparent, and operational approach that effectively bridges cutting-edge AI techniques with reliable, explainable decision-making in Indian financial markets.

Author Contribution M.P. was responsible for the conceptualization, methodology design, model development, data preprocessing, experimentation, visualization, and overall investigation. J.P. contributed to software implementation, validation testing, and assisted in data curation. A.S. provided supervision, project administration, and contributed to the review and editing of the manuscript. P.D. contributed to the literature review and assisted in the review and editing of the manuscript.

Data Availability The data used in this study is available from the corresponding author upon reasonable request.

Declarations

Ethics Approval Not applicable

Consent to Participate Not applicable

Consent for Publication Not applicable

Conflict of Interest The authors declare no competing interests.

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